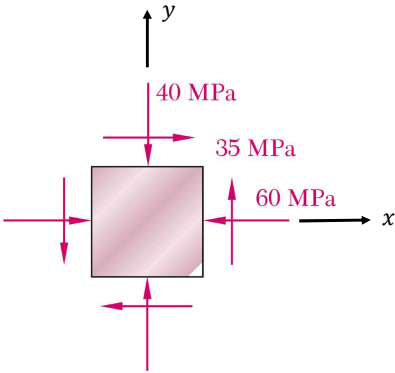


Zeshui Song, 4/20/26, HW9

7.31 For the given state of stress at left, determine (a) the principle planes and (b) the principle stresses, (c) the orientation of the planes of maximum in-plane shearing stress, and (d) the corresponding shearing and normal stresses. Draw Mohr's circle.



Given:

- $\sigma_x = -60 \text{ MPa}$
- $\sigma_y = -40 \text{ MPa}$
- $\tau_{xy} = 35 \text{ MPa}$

Find:

- Principle planes and stresses
- Orientation (angle) of the plane of max shear stress and corresponding shear and normal stresses

1

Zeshui Song, 4/20/26, HW9

Approach:

- Plot coordinates on Mohr's circle:

$$X: (\sigma_x, -\tau_{xy}), \quad Y: (\sigma_y, \tau_{xy}), \quad C: (\sigma_{ave}, 0)$$

- Define:

$$\sigma_{ave} = \frac{1}{2}(\sigma_x + \sigma_y), \quad \Delta\sigma = \sigma_x - \sigma_y, \quad R = \sqrt{\tau_{xy}^2 + \left(\frac{\Delta\sigma}{2}\right)^2}$$

- Find orientations for angle of principal planes and max shear:

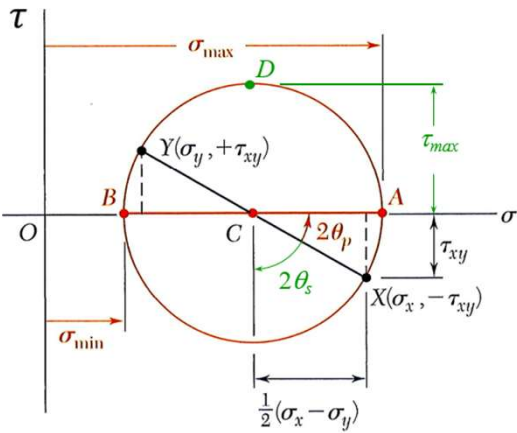
$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\Delta\sigma}, \quad \tan(2\theta_s) = -\frac{\Delta\sigma}{2\tau_{xy}}$$

- Principle normal stresses (max and min):

$$\sigma_{max} = \sigma_{ave} + R, \quad \sigma_{min} = \sigma_{ave} - R$$

- Max shear stress:

$$\tau_{max} = R$$



Note: the plane formed by XY represents the initial state of stress. To reach the max shear and principal stresses, we must rotate the orientation by angles described by θ_p & θ_s . (Since rotations in the Mohr circle represent twice the physical rotation, we use double angles)

2

Zeshui Song, 4/20/26, HW9

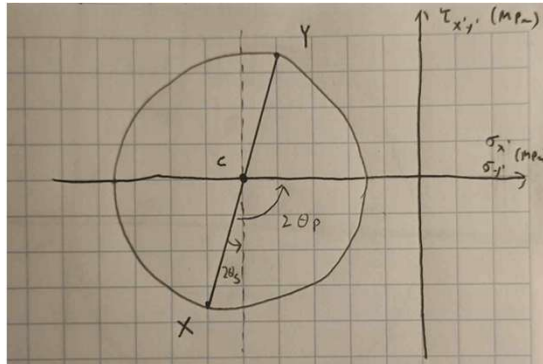
Plot coordinates on Mohr's circle:

$$\sigma_{ave} = \frac{1}{2}(\sigma_x + \sigma_y) = \frac{1}{2}[-60 - 40 \text{ MPa}] = -50 \text{ MPa}$$

$$\Delta\sigma = \sigma_x - \sigma_y = [-60 - (-40 \text{ MPa})] = -20 \text{ MPa}$$

$$X: (-60, -35), \quad Y: (-40, 35), \quad C: (-50, 0)$$

$$R = \sqrt{\tau_{xy}^2 + \left(\frac{\Delta\sigma}{2}\right)^2} = \sqrt{[35 \text{ MPa}]^2 + \left(\frac{[-20 \text{ MPa}]}{2}\right)^2} = 36.40 \text{ MPa}$$



3

Zeshui Song, 4/20/26, HW9

(a) Principal planes:

$$\theta_p = \frac{1}{2} \arctan \left(\frac{2\tau_{xy}}{\Delta\sigma} \right) = \frac{1}{2} \arctan \left(\frac{2[35 \text{ MPa}]}{[-20 \text{ MPa}]} \right) \quad \boxed{\theta_p = -37.027^\circ \text{ (CW)}}$$

(b) Principal stresses:

$$\sigma_{max} = \sigma_{ave} + R = [-50 \text{ MPa}] + [36.40 \text{ MPa}] = -13.6 \text{ MPa}$$

$$\sigma_{min} = \sigma_{ave} - R = [-50 \text{ MPa}] - [36.40 \text{ MPa}] = -86.4 \text{ MPa}$$

(c) Max shear plane:

$$\theta_s = \frac{1}{2} \arctan \left(-\frac{\Delta\sigma}{2\tau_{xy}} \right) = \frac{1}{2} \arctan \left(-\frac{[-20 \text{ MPa}]}{2[35 \text{ MPa}]} \right) \quad \boxed{\theta_s = 7.972^\circ \text{ (CCW)}}$$

(d) Max shear plane stresses:

$$\tau_{x'y'} = \tau_{max} = R = 36.40 \text{ MPa}$$

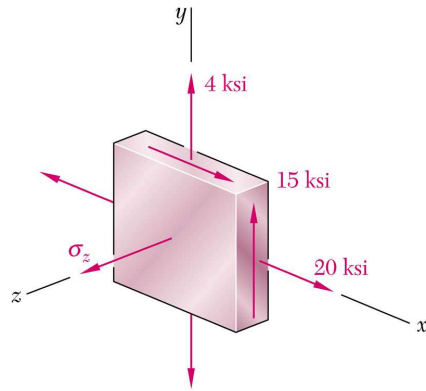
$$\sigma_{x'} = \sigma_{y'} = \sigma_{ave} = -50 \text{ MPa}$$

These results show that the critical (max) principal stresses are both compressive after rotating the physical orientation by θ_p . Additionally, when shear stress is max, both x and y faces sees equal amounts of compressive stress. [Time spent: 45 mins]

4

Zeshui Song, 4/20/26, HW9

7.70 For the state of stress shown at right, determine the maximum shearing stress when (a) $\sigma_z = 0$, (b) $\sigma_z = +9 \text{ ksi}$, and (c) $\sigma_z = -9 \text{ ksi}$. Discuss your results. (*Hint: Consider both in-plane and out-of-plane shearing stresses.*)



Given:

- $\sigma_x = 20 \text{ ksi}$
- $\sigma_y = 4 \text{ ksi}$
- $\tau_{xy} = 15 \text{ ksi}$

Find:

- Max shearing stress for 3 given possible σ_z values

5

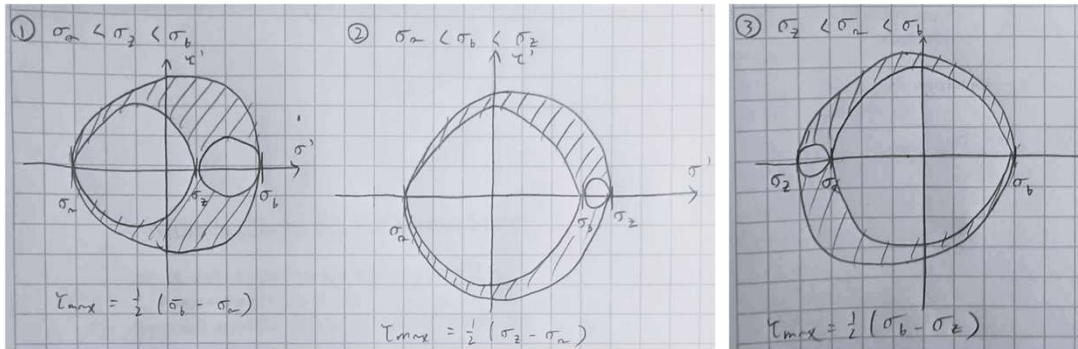
Zeshui Song, 4/20/26, HW9

Approach:

- This problem asks us to consider the max shearing stress for a 3d state of stress. In a 3D Mohr's circle, the max shear stress is given by:

$$\tau_{max} = R_{max} = \frac{\sigma_{max} - \sigma_{min}}{2}$$

- Since the 3D Mohr's circle is defined by the 3 principal stresses ($\sigma_a, \sigma_b, \sigma_c$). In-plane stress where $\sigma_z = 0$, we get two principal stresses from the max and min normal stresses. When $\sigma_z \neq 0$, it becomes the third principal stress.
- Thus, we can have 3 cases of 3D Mohr's circles depending on where σ_z is relative to the in-plane principal stresses.



Note: The possible stresses are defined by the boundaries of each Mohr's circle if we only rotate about ONE axis (in-plane stress). When we rotate about multiple axes, the possible stresses are defined by the shaded region in between (it is ALWAYS limited by the absolute max stresses defined by the outer circle)

6

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First finding the in-plane principal stresses

$$\sigma_{ave} = \frac{1}{2}(\sigma_x + \sigma_y) = \frac{1}{2}[20 + 4 \text{ ksi}] = 12 \text{ ksi}$$

$$\Delta\sigma = \sigma_x - \sigma_y = [20 - 4 \text{ ksi}] = 16 \text{ ksi}$$

$$R = \sqrt{\tau_{xy}^2 + \left(\frac{\Delta\sigma}{2}\right)^2} = \sqrt{[15 \text{ ksi}]^2 + \left(\frac{[16 \text{ ksi}]}{2}\right)^2} = 17 \text{ ksi}$$

$$\sigma_{max} = \sigma_{ave} + R = [12 \text{ ksi}] + [17 \text{ ksi}] = 29 \text{ ksi}$$

$$\sigma_{min} = \sigma_{ave} - R = [12 \text{ ksi}] - [17 \text{ ksi}] = -5 \text{ ksi}$$

$$\sigma_a = \sigma_{min} = -5 \text{ ksi}$$

$$\sigma_b = \sigma_{max} = 29 \text{ ksi}$$

(a) $\sigma_z = 0$: Case $\sigma_a < \sigma_z < \sigma_b$

$$\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2} = \frac{[29 - -5 \text{ ksi}]}{2}$$

$$\tau_{max} = 17 \text{ ksi}$$

(b) $\sigma_z \leq 9 \text{ ksi}$: Case $\sigma_a < \sigma_z < \sigma_b$

$$\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2} = \frac{[29 - -5 \text{ ksi}]}{2}$$

$$\tau_{max} = 17 \text{ ksi}$$

(c) $\sigma_z = -9$: Case $\sigma_z < \sigma_a < \sigma_b$

$$\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2} = \frac{[29 - -9 \text{ ksi}]}{2}$$

$$\tau_{max} = 19 \text{ ksi}$$

For scenarios where σ_z fell within the range of the in-plane principal stresses, the absolute max shear stress remained constant and was governed solely by the xy-plane. In Case (c), however, σ_z fell outside this range, becoming the new minimum principal stress. This resulted in a higher out-of-plane max shear stress. Failure can then occur in a different direction depending on the loading, which makes sense. [Time spent: 30 mins]

7

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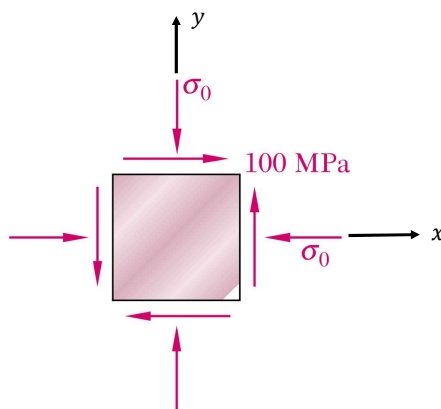
7.83 The state of plane stress shown at left occurs in a machine component made of a steel with $\sigma_Y = 325 \text{ MPa}$. Using the maximum-distortion-energy criterion, determine whether yield will occur when (a) $\sigma_0 = 200 \text{ MPa}$, (b) $\sigma_0 = 240 \text{ MPa}$, and (c) $\sigma_0 = 280 \text{ MPa}$. If yield does not occur, determine the corresponding factor of safety.

Given:

- $\sigma_x = \sigma_y = -\sigma_0$
- $\tau_{xy} = 100 \text{ MPa}$
- $\sigma_Y = 325 \text{ MPa}$

Find:

- For the three values σ_0 , determine if the material will yield. If not, find the factor of safety



8

Zeshui Song, 4/20/26, HW9

Approach:

- Find principal stresses using the Mohr's circle:

$$\sigma_a = \sigma_{max} = \sigma_{ave} + R, \quad \sigma_b = \sigma_{min} = \sigma_{ave} - R$$

- This problem asks for the von Mises Criterion (maximum distortion energy). Failure occurs when principal stresses fall outside the elliptical region:

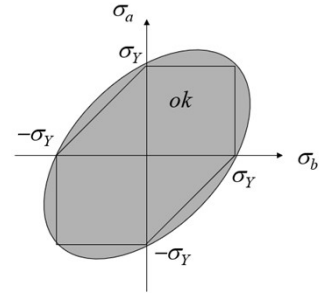
$$\sigma_a^2 - \sigma_a \sigma_b + \sigma_b^2 = \sigma_Y^2$$

In other words, the following inequality must be satisfied for it to not yield:

$$\sqrt{\sigma_a^2 - \sigma_a \sigma_b + \sigma_b^2} < \sigma_Y$$

- If the material does not yield, then the factor of safety is found by:

$$F.S. = \frac{\sigma_Y}{\sigma_{all}}$$



9

Zeshui Song, 4/20/26, HW9

First finding the in-plane principal stresses

$$\sigma_{ave} = \frac{1}{2}(\sigma_x + \sigma_y) = \frac{1}{2}(-2\sigma_0) = -\sigma_0$$

$$\Delta\sigma = \sigma_x - \sigma_y = \sigma_0 - \sigma_0 = 0$$

$$\sigma_{max} = \sigma_{ave} + R = -\sigma_0 + [100 \text{ MPa}]$$

$$\sigma_{min} = \sigma_{ave} - R = -\sigma_0 - [100 \text{ MPa}]$$

$$\sigma_a = -\sigma_0 + [100 \text{ MPa}]$$

$$\sigma_b = -\sigma_0 - [100 \text{ MPa}]$$

(a) $\sigma_0 = 200 \text{ MPa}$

$$\sigma_a = [-200 + 100 \text{ MPa}] = -100 \text{ MPa}$$

$$\sigma_b = [-200 - 100 \text{ MPa}] = -300 \text{ MPa}$$

$$\sqrt{\sigma_a^2 - \sigma_a \sigma_b + \sigma_b^2} < \sigma_Y$$

$$\sqrt{[-100 \text{ MPa}]^2 - [-100 \cdot -300 \text{ MPa}^2] + [-300 \text{ MPa}]^2} = [264.575 \text{ MPa}] < [325 \text{ MPa}] \quad \text{(no yielding)}$$

$$F.S. = \frac{[325 \text{ MPa}]}{[264.575 \text{ MPa}]} = 1.228$$

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(b) $\sigma_0 = 240 \text{ MPa}$

$$\sigma_a = [-240 + 100 \text{ MPa}] = -140 \text{ MPa}$$

$$\sigma_b = [-240 - 100 \text{ MPa}] = -340 \text{ MPa}$$

$$\sqrt{[-140 \text{ MPa}]^2 - [-140 \cdot -340 \text{ MPa}^2] + [-340 \text{ MPa}]^2} = [295.972 \text{ MPa}] < [325 \text{ MPa}] \quad (\text{no yielding})$$

$$F.S. = \frac{[325 \text{ MPa}]}{[295.972 \text{ MPa}]} = 1.098$$

(c) $\sigma_0 = 280 \text{ MPa}$

$$\sigma_a = [-280 + 100 \text{ MPa}] = -180 \text{ MPa}$$

$$\sigma_b = [-280 - 100 \text{ MPa}] = -380 \text{ MPa}$$

$$\sqrt{[-180 \text{ MPa}]^2 - [-180 \cdot -380 \text{ MPa}^2] + [-380 \text{ MPa}]^2} = [329.241 \text{ MPa}] > [325 \text{ MPa}] \quad (\text{yielding})$$

This makes sense, as we increase the compressive stress, there will reach a point where yielding occurs. Since the factor of safety for $\sigma_0 = 240 \text{ MPa}$ was very close to 1, it is also close to the stress where yielding will begin.
[Time spent: 25 mins]

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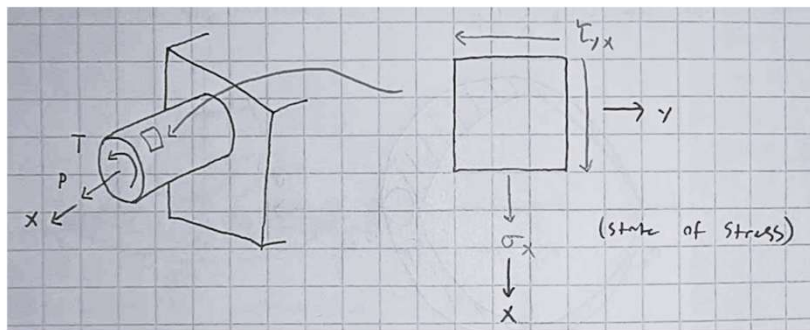
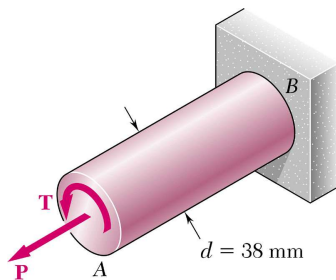
(GRADED) 7.87 The 38 mm diameter shaft AB at right is made of a grade of steel for which the yield strength is $\sigma_Y = 250 \text{ MPa}$. Using the maximum-shearing-stress criterion, determine the magnitude of the torque T for which yield occurs when $P = 240 \text{ kN}$.

Given:

- $\sigma_Y = 250 \text{ MPa}$
- $P = 240 \text{ kN}$
- $d = 38 \text{ mm}$

Find:

- Max torque for which yield occurs



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Approach:

- Define the state of stress on the surface of the shaft where the torsional shear stress is max. Where:

$$\sigma_x = \frac{P}{A} = \frac{P}{\frac{\pi}{4}d^2} = \frac{4P}{\pi d^2}, \quad \sigma_y = 0$$

$$\tau_{yx} = \frac{T(d/2)}{J}, \quad J = \frac{\pi}{2} \left(\frac{d}{2} \right)^4 = \frac{\pi}{32} d^4$$

- Apply the Tresca failure criterion (max shear stress) to relate the yield stress to the radius of Mohr's circle:

$$\tau_{max} = R = \frac{1}{2} \sigma_Y$$

- Use the radius equation for Mohr's circle to find max allowable torque:

$$R = \sqrt{\tau_{yx}^2 + \left(\frac{\Delta\sigma}{2} \right)^2}$$

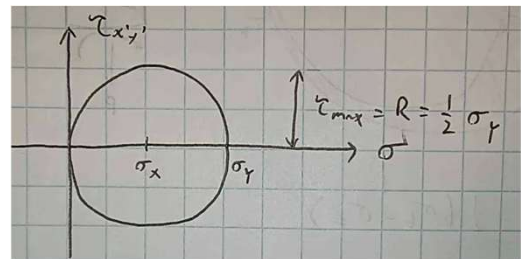
Note: This is derived from Mohr's circle for a uniaxial tensile test where:

$$\sigma_x = \frac{F}{A}, \quad \sigma_y = \sigma_z = 0, \quad \tau_{xy}, \tau_{xz}, \tau_{yz} = 0 \quad (\text{state of stress})$$

$$\sigma_{ave} = \sigma_x$$

Under ductile yield:

$$\sigma_{max} = \sigma_Y$$



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Define the state of stress

$$\sigma_x = \frac{4P}{\pi d^2} = \frac{4[240 \times 10^3 \text{ N}]}{\pi [0.038 \text{ m}]^2} = 211.618 \times 10^6 \text{ Pa} = 211.618 \text{ MPa} \quad \sigma_y = 0$$

$$\tau_{yx} = \frac{T(d/2)}{\frac{\pi}{32} d^4} = \frac{16T}{\pi d^3} = \frac{16}{\pi [0.038 \text{ m}]^3} T$$

Apply Tresca Criterion to find τ_{yx} when shear stress is max

$$\tau_{max} = \frac{1}{2} \sigma_Y = \frac{1}{2} [250 \text{ MPa}] = 125 \text{ MPa}$$

$$\tau_{max} = R = \sqrt{\tau_{yx}^2 + \left(\frac{\Delta\sigma}{2} \right)^2} = [125 \text{ MPa}] \quad \Delta\sigma = \sigma_x - \sigma_y = 211.618 \text{ MPa}$$

$$\sqrt{\tau_{yx}^2 + [105.809 \text{ MPa}]^2} = [125 \text{ MPa}] \rightarrow \boxed{\tau_{yx} = 66.554 \text{ MPa}}$$

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Use torsional stress equation for max allowable torque

$$\tau_{yx} = \frac{16T}{\pi d^3} \rightarrow T = \frac{\pi d^3 \tau_{yx}}{16}$$

$$T = \frac{\pi [0.038 \text{ m}]^3 [66.554 \times 10^6 \text{ Pa}]}{16} = 717.06 \text{ Nm}$$

By applying the Tresca failure criterion to the critical region of the shaft, we can determine the maximum allowable torque more accurately than by simply comparing the critical stress. As shown by Mohr's circle, the maximum shear stress includes contributions from both the normal and shear stresses in the state of stress.

Additionally, applying the Tresca failure criterion shows that the combination of normal and shear stress pushes the material closer to yielding. Since the load (P), and thus (σ_x), is so large, it contributes much more to yielding than the torque. [Time spent: 45 mins]

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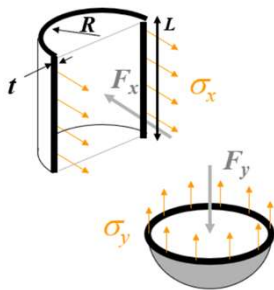
7.105 Determine the largest internal pressure that can be applied to a cylindrical tank of 1.75 m outer diameter and 16 mm wall thickness if the ultimate normal stress of the steel used is 450 MPa and a factor of safety of 5 is desired.

Approach:

- Use the factor of safety to find the allowable stress

$$F.S. = \frac{\sigma_U}{\sigma_{all}}$$

- The critical stress within a pressure vessel is the hoop stress, thus set hoop stress equal to allowable stress and solve for pressure

Hoop stress: σ_x , Axial stress: σ_y

$$\sum F_x = 0 = -P(L2R) + \sigma_x(2tL) \Rightarrow \sigma_x = \frac{PR}{t}$$

$$\sum F_y = 0 = -P(\pi R^2) + \sigma_y(\pi Rt) \Rightarrow \sigma_y = \frac{PR}{2t}$$

$$\Rightarrow \sigma_x = 2\sigma_y$$

Given:

- $d = 1.75 \text{ m}$
- $t = 16 \text{ mm}$
- $\sigma_U = 450 \text{ MPa}$
- $F.S. = 5$

Find:

- Max internal pressure

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Find allowable stress

$$\sigma_{all} = \frac{\sigma_U}{(F.S.)} = \frac{[450 \text{ MPa}]}{5} = 90 \text{ MPa}$$

Find pressure

$$R = \frac{d}{2} - t = \frac{[1.75 \text{ m}]}{2} - [0.016 \text{ m}] = 0.859 \text{ m}$$

$$\sigma_x = \sigma_{all} = \frac{PR}{t}$$

$$P = \frac{\sigma_{all} t}{R} = \frac{\left[90 \times 10^6 \frac{\text{N}}{\text{m}^2}\right] [0.016 \text{ m}]}{[0.859 \text{ m}]} = 1.676 \times 10^6 \text{ Pa}$$

Such a large factor of safety makes sense since when a pressure vessel fails, it could fail explosively and cause major damage. Thus, analyzing the critical stress at the hoop allows us to design a safe pressure. We don't need to look at the max shear stress from Mohr's circle since the factor of safety is only defined for normal stresses. [Time spent: 30 mins]

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Beam Bending Lab

Measure strain and deflection of a cantilever beam (see handout for procedure and datasheet). Sign up for a 1-hour time with Estuardo Rodas to run the lab in LL224, in teams of two students.

Lab Data Analysis: Calculation 1. Use your team/s dataset for strain vs. load to estimate the Young's modulus of your team's beam material. First, find the best fit slope of strain vs. force. Next derive an equation for the strain over the force (or $d\epsilon/dF$) in terms of the beam dimensions, strain gauge location, and beam material properties. Solve for the unknown Young's modulus using the values from your experiment. Check by comparing to published material properties and discuss briefly.

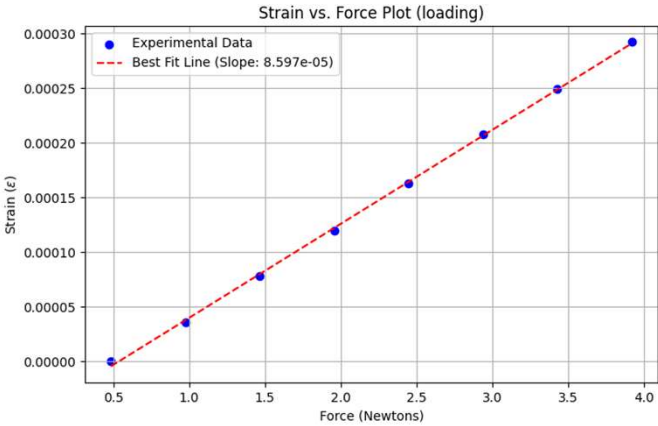
18

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Best fit slope of strain vs. force

- Using python, I will input the experimental data and plot a best fit line. Converting from cumulative load (g) to force (N) and microstrain to strain:

$$F = mg \rightarrow [N] = [g] \left[\frac{1 \text{ kg}}{10^3 \text{ g}} \right] \cdot \left[9.81 \frac{\text{m}}{\text{s}^2} \right]$$



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Deriving an equation for the strain over the force (or $d\epsilon/dF$)

Approach:

- Since the strain gauge is mounted on the upper surface of the beam, it sees zero shear stress and max tensile stress:

$$\sigma_{m,t} = \frac{M(h - c)}{I}$$

- Assuming the beam is in the linear elastic region, hooke's law relationship lets me rewrite normal stress in terms of strain:

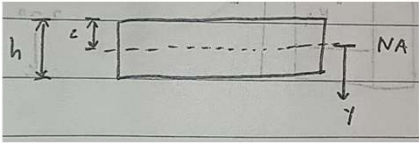
$$\sigma = E\epsilon$$

- Use a FBD to determine the moment at the position of the strain gauge based on beam dimensions
- Plug into the max tensile stress equation to solve for Young's Modulus based on $d\epsilon/dF$

Geometry

Given:

- $t = 0.1260$ in
- $L = 12.5$ in
- $w = 1.0020$ in
- Clamp to loading point $x_{c \rightarrow l} = 11.515$ in
- Loading point to micrometer $x_{l \rightarrow m} = 1.468$ in
- Clamp to strain gauge center $x_{c \rightarrow s} = 0.343$ in

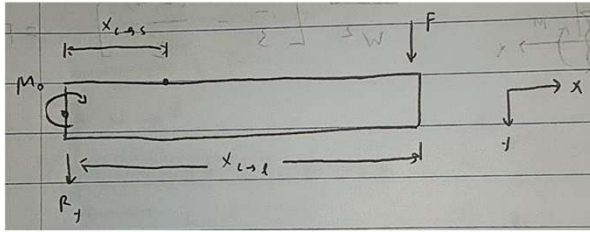


$$h - c = \frac{1}{2}t$$
$$I = \frac{1}{12}bh^3 = \frac{1}{12}wt^3$$

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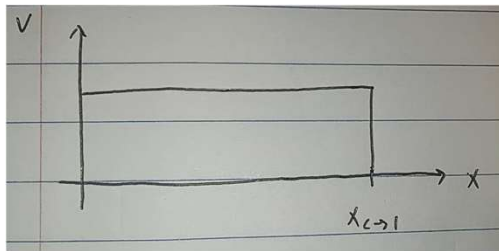
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FBD to determine stress distribution



Starting from the clamp, using the differential form:

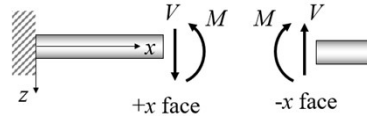
$$\frac{dV}{dx} = -w \rightarrow V = V(0) = F$$



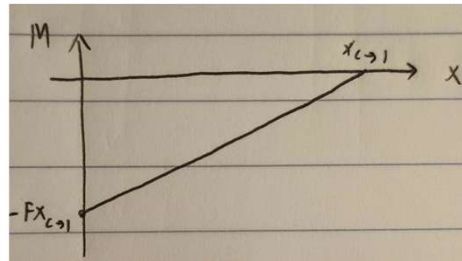
$$\sum F_y = 0 = R_y - F \rightarrow R_y = F$$

$$\sum M_z = 0 = M_0 - Fx_{c \rightarrow l} \rightarrow M_0 = Fx_{c \rightarrow l}$$

Note: sign convention



$$\frac{dM}{dx} = V \rightarrow M = M(0) + \int_0^x V dx' \rightarrow M = -Fx_{c \rightarrow l} + Fx$$



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Thus, the moment at the strain gauge ($x = x_{c \rightarrow s}$):

$$M = -Fx_{c \rightarrow l} + Fx_{c \rightarrow s} = |F(x_{c \rightarrow s} - x_{c \rightarrow l})|$$

Derive equation for Young's Modulus

$$\sigma_{m,t} = \frac{M(h-c)}{I} \rightarrow E\epsilon = \frac{M(h-c)}{I} \rightarrow E = \frac{M(h-c)}{I\epsilon}$$

Substitute in eqns:

$$E = \frac{|F(x_{c \rightarrow s} - x_{c \rightarrow l})|(t/2)}{\frac{1}{12}wt^3\epsilon} = \frac{6F|x_{c \rightarrow s} - x_{c \rightarrow l}|}{wt^2\epsilon}$$

Rewrite in terms of $d\epsilon/dF$

$$\epsilon = \frac{6F|x_{c \rightarrow s} - x_{c \rightarrow l}|}{wt^2E} \rightarrow \frac{d\epsilon}{dF} = \frac{6|x_{c \rightarrow s} - x_{c \rightarrow l}|}{wt^2E}$$

Solve for E :

$$E = \frac{6|x_{c \rightarrow s} - x_{c \rightarrow l}|}{wt^2 \left(\frac{d\epsilon}{dF} \right)}$$

$$E = \frac{6 \left[11.515 - 0.343 \text{ in} \cdot \frac{0.0254 \text{ m}}{1 \text{ in}} \right]}{\left[1.0020 \text{ in} \cdot \frac{0.0254 \text{ m}}{1 \text{ in}} \right] \left[0.1260 \text{ in} \cdot \frac{0.0254 \text{ m}}{1 \text{ in}} \right]^2 [8.597 \times 10^{-5} \text{ N}^{-1}]} = 7.597 \times 10^{10} \text{ N/m}^2$$

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The calculated Young's modulus is relatively close to the published value for Aluminum Alloy 2014-T6 (75 GPa). The strain vs. force plot showed a linear relationship, which confirms that our assumption of a linear elastic region and a Hooke stress-strain relationship is valid. Additionally, it was interesting that while the beam is not under pure bending due to the presence of shear stress, there exists only normal stress at the surface containing the strain gauge when the beam is bending.